

(14)

$$\bar{A} = \begin{bmatrix} 3 & x-2i & -yi \\ 3+2i & 0 & 1-2i \\ -4i & 1+xi & -1 \end{bmatrix} \quad 3 \times 3$$

$$(\bar{A})^t = \begin{bmatrix} 3 & 3+2i & -4i \\ x-2i & 0 & 1-2i \\ -yi & 1+xi & -1 \end{bmatrix} \quad 3 \times 3$$

For Hermitian Matrix,  $A = (\bar{A})^t$

$$\Rightarrow A = (\bar{A})^t$$

$$\Rightarrow \begin{bmatrix} 3 & x+2i & yi \\ 3-2i & 0 & 1+2i \\ 4i & 1-xi & -1 \end{bmatrix} = \begin{bmatrix} 3 & 3+2i & -4i \\ x-2i & 0 & 1+xi \\ -4i & 1-2i & -1 \end{bmatrix}$$

$$x+2i = 3+2i \quad \text{--- (1)}$$

$$1+2i = 1+xi \quad \text{--- (4)}$$

$$yi = -4i \quad \text{--- (2)}$$

$$1-xi = 1-2i \quad \text{--- (5)}$$

$$3-2i = x-2i \quad \text{--- (3)}$$

$$yi = -yi \quad \text{--- (6)}$$

From (1)

$$x+2i = 3+2i$$

$$x = 3$$

From (4)

$$1+2i = 1+xi$$

$$2i = xi$$

From (2)

$$2i = 3i$$

$$yi = -yi$$

$$2 = 3$$

$$yi + yi = 0$$

$$y(2i) = 0$$

$$2 = 3$$

$$y = 0$$

Your Success is our win (15)

From (5)

$$x-xi = x-2i$$

$$13x = x-2i$$

$$z = 3$$

From (3)

$$2-yi = x-2i$$

$$x = 3$$

Thus,  $x = 3$ ,  $y = 0$ ,  $z = 3$  E.P.

## Ex # 2.2

$$\text{If } A = \begin{bmatrix} 1+i & 2 & 3i \\ 4i & 5-i & 2+3i \\ 0 & 5 & 1-i \end{bmatrix} \text{ and } B = \begin{bmatrix} 2+i & 3-i & 4 \\ i & 0 & 5-i \\ 6+i & 2 & 2+3i \end{bmatrix}$$

Sol

(i)  $A+B$

$$\Rightarrow A+B = \begin{bmatrix} 1+i & 2 & 3i \\ 4i & 5-i & 2+3i \\ 0 & 5 & 1-i \end{bmatrix} + \begin{bmatrix} 2+i & 3-i & 4 \\ i & 0 & 5-i \\ 6+i & 2 & 2+3i \end{bmatrix}$$

3x3

3x3

$$\Rightarrow A+B = \begin{bmatrix} 1+i+2+i & 2+3-i & 3i+4 \\ 4i+i & 5-i+0 & 2+3i+5-i \\ 0+6+i & 5+2 & 1-i+2+3i \end{bmatrix}$$

3x3

$$\Rightarrow A \times B = \begin{bmatrix} 3+2i & 5-i & 3i+4 \\ 5i & 5-i & 7+2i \\ 6+i & 7 & 3+3i \end{bmatrix}$$

3x3

(16)

(ii)  $A - B$ 

$$\Rightarrow A - B = \begin{bmatrix} 1+i & 2 & 3i \\ 4i & 5-i & 2+3i \\ 0 & 5 & 1-i \end{bmatrix} - \begin{bmatrix} 2+i & 3-i & 4 \\ i & 0 & 5-i \\ 6+i & 2 & 2+3i \end{bmatrix}$$

$3 \times 3 \qquad \qquad \qquad 3 \times 3$

$$\Rightarrow A - B = \begin{bmatrix} 1+i-2-i & 2-3+i & 3i-4 \\ 4i-i & 5-i-0 & 2+3i-5+i \\ 0-6-i & 5-2 & 1-i-2-3i \end{bmatrix}$$

$3 \times 3$

$$\Rightarrow A - B = \begin{bmatrix} -1 & -1+i & -4+3i \\ 3i & 5-i & -3+4i \\ -6-i & 3 & -1-4i \end{bmatrix}$$

$3 \times 3$

(iii)  $2A - B$ 

$$\Rightarrow 2A = 2 \begin{bmatrix} 1+i & 2 & 3i \\ 4i & 5-i & 2+3i \\ 0 & 5 & 1-i \end{bmatrix}$$

$3 \times 3$

$$\Rightarrow 2A = \begin{bmatrix} 2+2i & 4 & 6i \\ 8i & 10-2i & 4+6i \\ 0 & 10 & 2-2i \end{bmatrix}$$

$3 \times 3$

$$\Rightarrow 2A - B = \begin{bmatrix} 2+2i & 4 & 6i \\ 8i & 10-2i & 4+6i \\ 0 & 10 & 2-2i \end{bmatrix} - \begin{bmatrix} 2+i & 3-i & 4 \\ i & 0 & 5-i \\ 6+i & 2 & 2+3i \end{bmatrix}$$

$3 \times 3 \qquad \qquad \qquad 3 \times 3$

(17)

$$\Rightarrow 2A - B = \begin{bmatrix} 2+2i-2-i & 4-3+i & 6i-4 \\ 8i-i & 10-2i & 4+6i-5+i \\ -6-i & 10-2 & 2-2i-2-3i \end{bmatrix}$$

$3 \times 3$

$$\Rightarrow 2A - B = \begin{bmatrix} -i & 1+i & 6i-4 \\ 7i & 10-2i & -1+7i \\ -6-i & 8 & -5i \end{bmatrix}$$

$3 \times 3$

(iv)  $3A + B + I$ 

$$\Rightarrow 3A + B + I = 3 \begin{bmatrix} 1+i & 2 & 3i \\ 4i & 5-i & 2+3i \\ 0 & 5 & 1-i \end{bmatrix} + \begin{bmatrix} 2+i & 3-i & 4 \\ i & 0 & 5-i \\ 6+i & 2 & 2+3i \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$3 \times 3 \qquad \qquad \qquad 3 \times 3 \qquad \qquad \qquad 3 \times 3$

$$\Rightarrow 3A + B + I = \begin{bmatrix} 3+3i & 6 & 9i \\ 12i & 15-3i & 6+9i \\ 0 & 15 & 3-3i \end{bmatrix} + \begin{bmatrix} 2+i & 3-i & 4 \\ i & 1 & 5-i \\ 6+i & 2 & 3+3i \end{bmatrix}$$

$3 \times 3 \qquad \qquad \qquad 3 \times 3$

$$\Rightarrow 3A + B + I = \begin{bmatrix} 3+3i+2+i & 6+3-i & 9i+4 \\ 12i+i & 15-3i+1 & 6+9i+5-i \\ 0+6+i & 15+2 & 3-3i+3+3i \end{bmatrix}$$

$3 \times 3$

$$\Rightarrow 3A + B + I = \begin{bmatrix} 6+4i & 9-i & 9i+4 \\ 13i & 16-3i & 11+8i \\ 6+i & 17 & 6 \end{bmatrix}$$

$3 \times 3$

(18)

Let  $A = \begin{bmatrix} 3 & -4 & 1 \\ 4 & 5 & 7 \\ -2 & -3 & 8 \end{bmatrix}$ ,  $B = \begin{bmatrix} 3 & 4 & -1 \\ 2 & -3 & 5 \end{bmatrix}$  &  $C = \begin{bmatrix} -5 & 4 \\ 6 & -3 \\ -2 & 7 \end{bmatrix}$

(i) AC

S.I

 $= AC$ 

$$= \begin{bmatrix} 3 & -4 & 1 \\ 4 & 5 & 7 \\ -2 & -3 & 8 \end{bmatrix}_{3 \times 3} \begin{bmatrix} -5 & 4 \\ 6 & -3 \\ -2 & 7 \end{bmatrix}_{3 \times 2}$$

$$= \begin{bmatrix} (3)(-5) + (-4)(6) + (1)(-2) & (3)(4) + (-4)(-3) + (1)(7) \\ (4)(-5) + (5)(6) + (7)(-2) & (4)(4) + (5)(-3) + (7)(7) \\ (-2)(-5) + (-3)(6) + (8)(-2) & (-2)(4) + (-3)(-3) + (8)(7) \end{bmatrix}_{3 \times 2}$$

$$= \begin{bmatrix} -15 - 24 - 2 & 12 + 12 + 7 \\ -20 + 30 - 14 & 16 - 15 + 49 \\ 10 - 18 - 16 & -8 + 9 + 56 \end{bmatrix}_{3 \times 2}$$

$$= \begin{bmatrix} -41 & 31 \\ -4 & 50 \\ -24 & 57 \end{bmatrix}_{3 \times 2} \quad \text{Ans}$$

(ii) BC

S.I

 $= BC$ 

$$= \begin{bmatrix} 3 & 4 & -1 \\ 2 & -3 & 5 \end{bmatrix}_{2 \times 3} \begin{bmatrix} -5 & 4 \\ 6 & -3 \\ -2 & 7 \end{bmatrix}_{3 \times 2}$$

$$= \begin{bmatrix} (3)(-5) + (4)(6) + (-1)(-2) & (3)(4) + (4)(-3) + (-1)(7) \\ (2)(-5) + (-3)(6) + (5)(-2) & (2)(4) + (-3)(-3) + (5)(7) \end{bmatrix}_{2 \times 2}$$

2x2

(19)

$$= \begin{bmatrix} -15 + 24 + 2 & 12 - 12 - 7 \\ -10 - 18 - 10 & 8 + 9 + 35 \end{bmatrix}_{2 \times 2}$$

$$= \begin{bmatrix} 11 & -7 \\ -38 & 52 \end{bmatrix}_{2 \times 2} \quad \text{Ans}$$

(iii) AB

S.I

 $= AB$ 

$$= \begin{bmatrix} 3 & -4 & 1 \\ 4 & 5 & 7 \\ -2 & -3 & 8 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 3 & 4 & -1 \\ 2 & -3 & 5 \end{bmatrix}_{2 \times 3}$$

= Does not exist

(iv) BA

S.I

 $= BA$ 

$$= \begin{bmatrix} 3 & 4 & -1 \\ 2 & -3 & 5 \end{bmatrix}_{2 \times 3} \begin{bmatrix} 3 & -4 & 1 \\ 4 & 5 & 7 \\ -2 & -3 & 8 \end{bmatrix}_{3 \times 3}$$

$$= \begin{bmatrix} (3)(3) + (4)(4) + (-1)(-2) & (3)(-4) + (4)(5) + (-1)(-3) & (3)(1) + (4)(7) + (-1)(8) \\ (2)(3) + (-3)(4) + (5)(-2) & (2)(-4) + (-3)(5) + (5)(-3) & (2)(1) + (-3)(7) + (5)(8) \end{bmatrix}_{2 \times 3}$$

$$= \begin{bmatrix} 9 + 16 + 2 & -12 + 20 + 3 & 3 + 28 - 8 \\ 6 - 12 - 10 & -8 - 15 - 15 & 2 - 21 + 40 \end{bmatrix}_{2 \times 3}$$

$$= \begin{bmatrix} 27 & 11 & 23 \\ -16 & -32 & 21 \end{bmatrix}_{2 \times 3}$$

2x3

(20)

$$vi) A^2$$

$$\underline{S.1}$$

$$= A^2$$

$$= A \cdot A$$

$$= \begin{bmatrix} 3 & -4 & 1 \\ 4 & 5 & 7 \\ -2 & -3 & 8 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 3 & -4 & 1 \\ 4 & 5 & 7 \\ -2 & -3 & 8 \end{bmatrix}_{3 \times 3}$$

$$= \begin{bmatrix} (3)(3) + (-4)(4) + (1)(-2) & (3)(-4) + (-4)(5) + (1)(-3) & (3)(1) + (-4)(7) + (1)(8) \\ (4)(3) + (5)(4) + (7)(-2) & (4)(-4) + (5)(5) + (7)(-3) & (4)(1) + (5)(7) + (7)(8) \\ (-2)(3) + (-3)(4) + (8)(-2) & (-2)(-4) + (-3)(5) + (8)(-3) & (-2)(1) + (-3)(7) + (8)(8) \end{bmatrix}$$

$$= \begin{bmatrix} 9-16-2 & -12-20-3 & 3-28+8 \\ 12+20-14 & -16+25-21 & 4+35+56 \\ -6-12-16 & 8-15-24 & -2-21+64 \end{bmatrix}_{3 \times 3}$$

$$= \begin{bmatrix} -9 & -35 & -17 \\ 18 & -12 & 95 \\ -34 & -31 & 41 \end{bmatrix}_{3 \times 3} \text{ A.P.}$$

$$vii) CB$$

$$\underline{S.1}$$

$$= CB$$

$$= \begin{bmatrix} -5 & 4 \\ 6 & -3 \\ -2 & 7 \end{bmatrix}_{3 \times 2} \begin{bmatrix} 3 & 4 & -1 \\ 2 & -3 & 5 \end{bmatrix}_{2 \times 3}$$

$$= \begin{bmatrix} (-5)(3) + (4)(2) & (-5)(4) + (4)(-3) & (-5)(-1) + (4)(5) \\ (6)(3) + (-3)(2) & (6)(4) + (-3)(-3) & (6)(-1) + (-3)(5) \\ (-2)(3) + (7)(2) & (-2)(4) + (7)(-3) & (-2)(-1) + (7)(5) \end{bmatrix}_{3 \times 3}$$

(21)

$$= \begin{bmatrix} -15+8 & -20-12 & 5+20 \\ 18-6 & 24+9 & -6-15 \\ -6+14 & -8-21 & 2+35 \end{bmatrix}_{3 \times 3}$$

$$= \begin{bmatrix} -7 & -32 & 25 \\ 12 & 33 & -21 \\ 8 & -29 & 37 \end{bmatrix}_{3 \times 3} \text{ A.P.}$$

$$viii) (AB)C$$

$$\underline{S.1}$$

$$= (AB)C$$

First we find AB

$$= AB$$

$$= \begin{bmatrix} 3 & -4 & 1 \\ 4 & 5 & 7 \\ -2 & -3 & 8 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 3 & 4 & -1 \\ 2 & -3 & 5 \end{bmatrix}_{2 \times 3}$$

$\Rightarrow$  does not exist A.P.

$$ix) C^t B^t$$

$$\underline{S.1}$$

$$= C^t B^t$$

$$\Rightarrow C^t = \begin{bmatrix} -5 & 6 & -2 \\ 4 & -3 & 7 \end{bmatrix}_{2 \times 3}$$

$$\Rightarrow B^t = \begin{bmatrix} 3 & 2 \\ 4 & -3 \\ -1 & 5 \end{bmatrix}_{3 \times 2}$$

Now,

(22)

$$\begin{aligned}
 &= \begin{bmatrix} -5 & 6 & -2 \\ 4 & -3 & 7 \end{bmatrix}_{2 \times 3} \begin{bmatrix} 3 & 2 \\ 4 & -3 \\ -1 & 5 \end{bmatrix}_{3 \times 2} \\
 &= \begin{bmatrix} (-5)(3) + (6)(4) + (-2)(-1) & (-5)(2) + (6)(-3) + (-2)(5) \\ (4)(3) + (-3)(4) + (7)(-1) & (4)(2) + (-3)(-3) + (7)(5) \end{bmatrix}_{2 \times 2} \\
 &= \begin{bmatrix} -15 + 24 + 2 & -10 - 18 - 10 \\ 12 - 12 - 7 & 8 + 9 + 35 \end{bmatrix}_{2 \times 2} \\
 &= \begin{bmatrix} 11 & -38 \\ -7 & 52 \end{bmatrix}_{2 \times 2} \text{ ep}
 \end{aligned}$$

(ix)  $C^t A^t$ 

Sol

$$= C^t A^t$$

$$\Rightarrow C^t = \begin{bmatrix} -5 & 6 & -2 \\ 4 & -3 & 7 \end{bmatrix}_{2 \times 3}$$

$$\Rightarrow A^t = \begin{bmatrix} 3 & 4 & -2 \\ -4 & 5 & -3 \\ 1 & 7 & 8 \end{bmatrix}_{3 \times 3}$$

Now,

$$= \begin{bmatrix} -5 & 6 & -2 \\ 4 & -3 & 7 \end{bmatrix}_{2 \times 3} \begin{bmatrix} 3 & 4 & -2 \\ -4 & 5 & -3 \\ 1 & 7 & 8 \end{bmatrix}_{3 \times 3}$$

$$= \begin{bmatrix} (-5)(3) + (6)(-4) + (-2)(1) & (-5)(4) + (6)(5) + (-2)(7) & (-5)(-2) + (6)(-3) + (-2)(8) \\ (4)(3) + (-3)(-4) + (7)(1) & (4)(4) + (-3)(5) + (7)(7) & (4)(-2) + (-3)(-3) + (7)(8) \end{bmatrix}$$

$$= \begin{bmatrix} -15 - 24 - 2 & -20 + 30 - 14 & 10 - 18 - 16 \\ 12 + 12 + 7 & 16 - 15 + 49 & -8 + 9 + 56 \end{bmatrix}_{2 \times 3}$$

$$= \begin{bmatrix} -41 & -4 & -24 \\ 31 & 50 & 57 \end{bmatrix}_{2 \times 3} \text{ ep}$$

(23)

(x)  $(CA)B$ 

Sol

$$= (CA)B$$

= First we find CA

$$= \begin{bmatrix} -5 & 4 \\ 6 & -3 \\ -2 & 7 \end{bmatrix}_{3 \times 2} \begin{bmatrix} 3 & -4 & 1 \\ 4 & 5 & 7 \\ -2 & -3 & 8 \end{bmatrix}_{3 \times 3}$$

= Does not exist

$$(5) \text{ let, } X = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & -2 \\ 4 & -3 & -1 \end{bmatrix}, Y = \begin{bmatrix} 2 & 1 & -1 \\ 3 & -2 & -1 \\ 2 & -5 & -1 \end{bmatrix} \text{ \& } Z = \begin{bmatrix} 1 & 4 & 1 \\ 2 & 1 & 1 \\ 1 & -2 & 1 \end{bmatrix}$$

then show that  $XY = XZ$ 

Take L.H.S

$$= XY$$

$$= \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & -2 \\ 4 & -3 & -1 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 2 & 1 & -1 \\ 3 & -2 & -1 \\ 2 & -5 & -1 \end{bmatrix}_{3 \times 3}$$

$$= \begin{bmatrix} (1)(2) + (-3)(3) + (2)(2) & (2)(1) + (-3)(-2) + (2)(-5) & (1)(-1) + (-3)(-1) + (2)(-1) \\ (2)(2) + (1)(3) + (-2)(2) & (2)(1) + (1)(-2) + (-2)(-5) & (2)(-1) + (1)(-1) + (-2)(-1) \\ (4)(2) + (-3)(3) + (-1)(2) & (4)(1) + (-3)(-2) + (-1)(-5) & (4)(-1) + (-2)(-1) + (-1)(-1) \end{bmatrix}_{3 \times 3}$$

$$= \begin{bmatrix} 2 - 9 + 4 & 2 + 6 - 10 & -1 + 3 - 2 \\ 4 + 3 - 6 & 2 - 2 + 10 & -2 - 1 + 2 \\ 8 - 9 - 2 & 4 + 6 + 5 & -4 + 2 + 1 \end{bmatrix}_{3 \times 3}$$

$$= \begin{bmatrix} -3 & -2 & 0 \\ 1 & 8 & -1 \\ -3 & 15 & -1 \end{bmatrix}_{3 \times 3}$$

24

Take R.H.S

$= XZ$

$$= \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & -3 \\ 4 & -3 & -1 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 1 & 4 & 1 \\ 2 & 1 & 1 \\ 1 & -2 & 1 \end{bmatrix}_{3 \times 3}$$

$$= \begin{bmatrix} (1)(1) + (-3)(2) + (2)(1) & (1)(4) + (-3)(1) + (2)(-2) & (1)(1) + (-3)(1) + (2)(1) \\ (2)(1) + (1)(2) + (-3)(1) & (2)(4) + (1)(1) + (-3)(-2) & (2)(1) + (1)(1) + (-3)(1) \\ (4)(1) + (-3)(2) + (-1)(1) & (4)(4) + (-3)(1) + (-1)(-2) & (4)(1) + (-3)(1) + (-1)(1) \end{bmatrix}_{3 \times 3}$$

$$= \begin{bmatrix} 1-6+2 & 4-3-4 & 1-3+2 \\ 2+2-3 & 8+1+6 & 2+1-3 \\ 4-6-1 & 16-3+2 & 4-3-1 \end{bmatrix}_{3 \times 3}$$

$$= \begin{bmatrix} -3 & -3 & 0 \\ 1 & 15 & 0 \\ -3 & 15 & 0 \end{bmatrix}_{3 \times 3}$$

Hence L.H.S = R.H.S

④ If  $A = \begin{bmatrix} -2 & 1 & 0 \\ -1 & 4 & 3 \\ 0 & 8 & 5 \end{bmatrix}$  then find  $A^2 - 5A + 4I$

First we find  $A^2$ .

$$A^2 = A \cdot A$$

$$= \begin{bmatrix} -2 & 1 & 0 \\ -1 & 4 & 3 \\ 0 & 8 & 5 \end{bmatrix}_{3 \times 3} \begin{bmatrix} -2 & 1 & 0 \\ -1 & 4 & 3 \\ 0 & 8 & 5 \end{bmatrix}_{3 \times 3}$$

$$= \begin{bmatrix} (-2)(-2) + (1)(-1) + (0)(0) & (-2)(1) + (1)(4) + (0)(8) & (-2)(0) + (1)(3) + (0)(5) \\ (-1)(-2) + (4)(-1) + (3)(0) & (-1)(1) + (4)(4) + (3)(8) & (-1)(0) + (4)(3) + (3)(5) \\ (0)(-2) + (8)(-1) + (5)(0) & (0)(1) + (8)(4) + (5)(8) & (0)(0) + (8)(3) + (5)(5) \end{bmatrix}_{3 \times 3}$$

25

$$A^2 = \begin{bmatrix} 4 & -1 & 3 \\ 2 & -4 & 12 \\ -8 & 32 & 24 \end{bmatrix}_{3 \times 3}$$

$$= A^2 = \begin{bmatrix} 3 & 2 & 3 \\ -2 & 39 & 27 \\ -8 & 72 & 49 \end{bmatrix}_{3 \times 3}$$

Now,

$$A^2 - 5A + 4I = \begin{bmatrix} 3 & 2 & 3 \\ -2 & 39 & 27 \\ -8 & 72 & 49 \end{bmatrix}_{3 \times 3} - 5 \begin{bmatrix} -2 & 1 & 0 \\ -1 & 4 & 3 \\ 0 & 8 & 5 \end{bmatrix}_{3 \times 3} + 4 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3 \times 3}$$

$$= A^2 - 5A + 4I = \begin{bmatrix} 3 & 2 & 3 \\ -2 & 39 & 27 \\ -8 & 72 & 49 \end{bmatrix}_{3 \times 3} - \begin{bmatrix} -10 & 5 & 0 \\ -5 & 20 & 15 \\ 0 & 40 & 25 \end{bmatrix}_{3 \times 3} + \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}_{3 \times 3}$$

$$= A^2 - 5A + 4I = \begin{bmatrix} 3+10 & 2-5 & 3-0 \\ -2+5 & 39-20 & 27-15 \\ -8+0 & 72-40 & 49-25 \end{bmatrix}_{3 \times 3} + \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}_{3 \times 3}$$

$$= A^2 - 5A + 4I = \begin{bmatrix} 13 & -3 & 3 \\ 3 & 19 & 12 \\ -8 & 32 & 24 \end{bmatrix}_{3 \times 3} + \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}_{3 \times 3}$$

$$= A^2 - 5A + 4I = \begin{bmatrix} 17 & -3 & 3 \\ 3 & 19 & 12 \\ -8 & 32 & 28 \end{bmatrix}_{3 \times 3}$$

$$= A^2 - 5A + 4I = \begin{bmatrix} 17 & -3 & 3 \\ 3 & 19 & 12 \\ -8 & 32 & 28 \end{bmatrix}_{3 \times 3}$$

26

5) Prove that:  $\begin{bmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \end{bmatrix} + \begin{bmatrix} \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \\ \omega & \omega^2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ \omega \\ \omega^2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Take L.H.S

$$= \begin{bmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \end{bmatrix}_{3 \times 3} + \begin{bmatrix} \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \\ \omega & \omega^2 & 1 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 1 \\ \omega \\ \omega^2 \end{bmatrix}_{3 \times 1}$$

$$= \begin{bmatrix} 1+\omega & \omega+\omega^2 & 1+\omega^2 \\ \omega+\omega^2 & 1+\omega^2 & 1+\omega \\ \omega+\omega^2 & 1+\omega^2 & 1+\omega \end{bmatrix} \begin{bmatrix} 1 \\ \omega \\ \omega^2 \end{bmatrix}$$

$$= \begin{bmatrix} 1+\omega+\omega^2+\omega^2+\omega^2+\omega^3 \\ \omega+\omega^2+\omega+\omega^3+\omega^2+\omega^2 \\ \omega+\omega^2+\omega+\omega^2+\omega^2+\omega^3 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 1 \\ \omega \\ \omega^2 \end{bmatrix}$$

$\because \omega^3 = 1$   
 $\therefore \omega^4 = \omega^3 \cdot \omega = 1 \cdot \omega = \omega$

$$= \begin{bmatrix} 1+\omega+\omega^2+1+\omega^2+\omega \\ \omega+\omega^2+\omega+1+\omega^2+1 \\ \omega+\omega^2+\omega+1+\omega^2+1 \end{bmatrix}_{3 \times 3}$$

$$= \begin{bmatrix} 1+\omega+\omega^2+1+\omega+\omega^2 \\ 1+\omega+\omega^2+1+\omega+\omega^2 \\ 1+\omega+\omega^2+1+\omega+\omega^2 \end{bmatrix}_{3 \times 3}$$

$\because 1+\omega+\omega^2 = 0$

$$= \begin{bmatrix} 0+0 \\ 0+0 \\ 0+0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

= R.H.S proved!

27

1) If  $A = \begin{bmatrix} 3 & 2 & 1 \\ 4 & -2 & 5 \\ -1 & 0 & -7 \end{bmatrix}$  and  $B = \begin{bmatrix} 9 & 4 & 6 \\ 7 & -8 & 5 \\ -1 & 0 & 3 \end{bmatrix}$  then verify:

8)  $A+B = B+A$

Take L.H.S

$$= A+B = \begin{bmatrix} 3 & 2 & 1 \\ 4 & -2 & 5 \\ -1 & 0 & -7 \end{bmatrix}_{3 \times 3} + \begin{bmatrix} 9 & 4 & 6 \\ 7 & -8 & 5 \\ -1 & 0 & 3 \end{bmatrix}_{3 \times 3}$$

$$= \begin{bmatrix} 3+9 & 2+4 & 1+6 \\ 4+7 & -2+8 & 5+5 \\ -1-1 & 0+0 & -7+3 \end{bmatrix}_{3 \times 3}$$

$$= \begin{bmatrix} 12 & 6 & 7 \\ 11 & -6 & 10 \\ -2 & 0 & -4 \end{bmatrix}_{3 \times 3}$$

Take R.H.S

$$= B+A = \begin{bmatrix} 9 & 4 & 6 \\ 7 & -8 & 5 \\ -1 & 0 & 3 \end{bmatrix}_{3 \times 3} + \begin{bmatrix} 3 & 2 & 1 \\ 4 & -2 & 5 \\ -1 & 0 & -7 \end{bmatrix}_{3 \times 3}$$

$$= \begin{bmatrix} 9+3 & 4+2 & 6+1 \\ 7+4 & -8-2 & 5+5 \\ -1-1 & 0+0 & 3-7 \end{bmatrix}_{3 \times 3}$$

$$= \begin{bmatrix} 12 & 6 & 7 \\ 11 & -10 & 10 \\ -2 & 0 & -4 \end{bmatrix}_{3 \times 3}$$

Hence, L.H.S = R.H.S proved!

(28)

$$(i) (AB)^t = B^t A^t$$

Take L.H.S

$$(AB)^t$$

Find we find AB

$$\Rightarrow AB = \begin{bmatrix} 7 & 2 & 1 \\ 4 & -2 & 5 \\ -1 & 0 & -7 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 3 & 4 & -1 \\ 2 & -2 & 0 \\ 1 & 5 & -7 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow AB = \begin{bmatrix} (1)(7) + (2)(2) + (1)(1) & (1)(4) + (2)(-2) + (1)(0) & (1)(-1) + (2)(5) + (1)(-7) \\ (2)(7) + (-2)(2) + (0)(1) & (2)(4) + (-2)(-2) + (0)(0) & (2)(-1) + (-2)(5) + (0)(-7) \\ (3)(7) + (0)(2) + (-7)(1) & (3)(4) + (0)(-2) + (-7)(0) & (3)(-1) + (0)(5) + (-7)(-7) \end{bmatrix}$$

$$\Rightarrow AB = \begin{bmatrix} 22+4-1 & 12-4+0 & -1+10-7 \\ 14-4+0 & 16+4+0 & -2-10+0 \\ 21+0-7 & 12+0-7 & -3+0+49 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow AB = \begin{bmatrix} 25 & 8 & 2 \\ 10 & 20 & -12 \\ 14 & 12 & 46 \end{bmatrix}_{3 \times 3}$$

$$(AB)^t$$

$$\Rightarrow \begin{bmatrix} 25 & 10 & 14 \\ 8 & 20 & 12 \\ 2 & -12 & 46 \end{bmatrix}_{3 \times 3}$$

Take R.H.S

$$A^t = \begin{bmatrix} 3 & 4 & -1 \\ 2 & -2 & 0 \\ 1 & 5 & -7 \end{bmatrix}_{3 \times 3}$$

$$B^t = \begin{bmatrix} 3 & 4 & -1 \\ 2 & -2 & 0 \\ 1 & 5 & -7 \end{bmatrix}_{3 \times 3}$$

Your Success is our win

(29)

$$B^t A^t$$

$$\Rightarrow \begin{bmatrix} 3 & 4 & -1 \\ 2 & -2 & 0 \\ 1 & 5 & -7 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 3 & 4 & -1 \\ 2 & -2 & 0 \\ 1 & 5 & -7 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow \begin{bmatrix} (1)(3) + (2)(2) + (1)(1) & (1)(4) + (2)(-2) + (1)(0) & (1)(-1) + (2)(0) + (1)(-7) \\ (2)(3) + (-2)(2) + (0)(1) & (2)(4) + (-2)(-2) + (0)(0) & (2)(-1) + (-2)(0) + (0)(-7) \\ (3)(3) + (0)(2) + (-7)(1) & (3)(4) + (0)(-2) + (-7)(0) & (3)(-1) + (0)(0) + (-7)(-7) \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 12-4-1 & 16-4+0 & -1+0-7 \\ 6-4+0 & 16+4+0 & -2+0+0 \\ 9+0-7 & 12+0-7 & -3+0+49 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow \begin{bmatrix} 7 & 12 & -8 \\ 2 & 20 & -2 \\ 2 & 12 & 46 \end{bmatrix}_{3 \times 3}$$

Ex # 2.3

Evaluate the following determinant.

$$(i) \begin{vmatrix} 5 & 4 & 3 \\ 3 & -4 & 0 \\ 2 & 3 & 1 \end{vmatrix}$$

$$\begin{vmatrix} 5 & 4 & 3 \\ 3 & -4 & 0 \\ 2 & 3 & 1 \end{vmatrix}$$

$$\text{Let } |D| = \begin{vmatrix} 5 & 4 & 3 \\ 3 & -4 & 0 \\ 2 & 3 & 1 \end{vmatrix}$$

$$|D| = 5 \begin{vmatrix} -4 & 0 \\ 3 & 1 \end{vmatrix} - 4 \begin{vmatrix} 3 & 0 \\ 2 & 1 \end{vmatrix} + 3 \begin{vmatrix} 3 & -4 \\ 2 & 3 \end{vmatrix}$$

$$|D| = 5[(-4)(1) - (0)(3)] - 4[(3)(1) - (0)(2)] + 3[(3)(3) - (-4)(2)]$$

$$|D| = 5(-4-0) - 4(3-0) + 3(9+8)$$